



An Intelligent Decision-Support Framework for Sustainable Logistics Hub Prioritization under Uncertainty: A q -Rung Orthopair Fuzzy OPA Approach

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Abstract: Sustainable logistics hub planning in emerging economies is often challenged by high levels of uncertainty, limited data availability, and the need to balance economic, environmental, and social objectives. Supporting consistent and transparent decision-making under such conditions remains a key issue in infrastructure planning. To address this, the present study develops an intelligent decision-support framework for prioritizing logistics hubs in complex and uncertain environments. The proposed framework combines q -rung orthopair fuzzy sets with the ordinal priority approach, enabling the representation of imprecise expert judgments alongside ordinal preference information within a unified multi-criteria structure. The approach is applied to the case of Kenya, where logistics development involves multiple and often conflicting criteria. A comprehensive evaluation system is established, and expert assessments are incorporated to derive priority rankings. The results show that operational efficiency and economic considerations play a dominant role in the decision process, while environmental and social factors receive comparatively lower weights. Sensitivity and comparative analyses confirm the stability and reliability of the findings. The study provides a structured and uncertainty-aware decision-support tool that can assist infrastructure planning and offers practical insights for policy and managerial decision-making in logistics systems.

Keywords: Sustainable logistics hubs; Ordinal priority approach; q -rung orthopair fuzzy sets; Infrastructure planning; Regional integration; East Africa; Kenya

1 Introduction

In the last several decades, the logistics hub’s way of planning and developing has been the subject of major changes. The traditional model of infrastructure planning and site selection was mostly about cutting costs and boosting operational capacity. Nowadays, the increasing number of policymakers deals with a possibility that the economic efficiency is not enough [1]. Big down turns in the global economy like the financial crisis, pandemic COVID-19, and high levels of climate pressure have shown the previous approaches are needful of considering social equity, some of their consequences are environmental degradation even in some areas.

The logistics, as of today, is responsible for about 5.5% of the greenhouse gas emissions globally, while the proportion of logistics buildings is considerable too (13%) [2, 3]. In Europe, the situation is much the same, as goods delivery and transshipment facilities contribute up to 30% of transportation emissions, depending on the region [4]. In the case of Germany, logistic real estate makes up around 15–50% of the emissions from the corporations operating in the country. The numbers are indeed illustrative of the fact that, on the one hand, logistics centers are important for trade and, on the other hand, they are a crucial point in driving the fulfillment of climate change mitigation goals [5–7].

These points are complicated even further by the rapid urbanization, inadequate infrastructure, and East African extreme climate change vulnerability. In the past few years, the geopolitics of the region has forced the making of Soft and Hard infrastructure, which includes the construction of roads and railroads as well as telecommunications.

However, Kenya's Standard Gauge Railway and Lamu Port-South Sudan-Ethiopia Transport Corridor (LAPSSET) are new projects that not only create alternatives to regional infrastructure bottlenecks but also promote development. Detractors had the overall picture of the damage to environment sustainability and the displacement of locals that could have arisen [8]. These challenges raise the case for being creative in deciding the area for logistics infrastructure to be developed and in the alternative ways logistics sites are to be planned. The transportation sector has the major environmental footprint now. Environmental protection is not the only thing. Promoting social equity must also be taken into consideration in the decision-making process. The complicated problems that this or the other decision-making models do not manage well. Some common multi-criteria decision-making methods (MCDM), for instance, the Analytical Hierarchy Process (AHP) and TOPSIS, are good evaluation frameworks. Owing to their ability to avoid problems like too many criteria being involved or data being limited, however, they can result in inconsistent outcomes when so many criteria are involved or data is scarce [9, 10]. The same constraints have influenced the development of new ways that are flexible enough to handle the hazards. One of the models is the q -ranked orthopair fuzzy set-based ordinal priority approach (q -ROFS-based OPA). This mechanism functions on ranking-based preference relationships. Therefore, it is made simpler for the decision-makers to communicate their points of view. Besides that, this approach affirms its high analytical reliability with the consistent and logically connected weighting among the criteria [11–14]. This research investigates the placement of the logistics hub in Kenya through the q -ROFS-based OPA. The assessment encompasses the five key dimensions of sustainability: environmental, economic, social, logistics/operational, and strategic/geopolitical factors). This study has three objectives: first, to identify a comprehensive set of sustainability criteria suited to logistics hubs in East Africa; second, to apply the q -ROFS-based OPA method to determine the relative importance of these criteria based on the input from experts; and third, to come up with policy recommendations for governments, investors, and development partners involved in infrastructure planning.

2 Literature Review

2.1 From Efficiency to Sustainability in Global Supply Chains

Today, logistics planning increasingly draws on the triple bottom framework—balancing economic, environmental, and social performance.

The traditional metrics of throughput and capacity have given way to a sharper focus on environmental footprint when evaluating and planning for logistics hubs. Warehousing, for example, might only account for 20% total logistics costs in many supply chains, yet it contributes to a disproportionately high emission [3]. Research from Europe puts logistics real estate at 11–30% of transport-related emissions, depending on the country and setup [4].

Even though warehousing accounts for only about 20% of total logistics costs, research shows that it contributes to a disproportionately high emission [3]. Studies from across Europe have shown that logistics real estate accounts for between 11% and 30% of all logistics-related emissions [4], with embodied carbon within these facilities contributing several tons of carbon dioxide throughout the life of the building [15]. The electricity consumed in the running of these facilities and refrigerant use only adds to the amount of emissions. These findings show the level of significance logistics infrastructure projects should put on the various aspects of the development, such as integrating renewable energy, eco-design, and waste reduction strategies in planning for the lifecycle of the project [7].

2.2 The East African Logistics Landscape

East Africa has an extensive geographical potential, but operationally this hasn't been the case. It is easier to carry cargo from overseas to the Port of Mombasa, however, it is impossible to efficiently push it inland to regions such as Kampala or Kigali. Poor road network, slow border procedures, and frequent port congestion have already delayed the shipment, and that is a norm for most operators. East African states have tried to unlock such bottlenecks through mega-projects like the SGR and LAPSSET corridor, but they seem to have locked in new concerns, particularly around environmental and social costs. The development of these large infrastructures has been associated with growing carbon footprints, loss of biodiversity, and displacement of communities [8]. When such promises turn into concerns, it becomes easier to question agreements such as AfCFTA claiming the unlocking of trillions of dollars of the continent's economic potential. The overriding question is not just about how fast and cheaply you move your goods. It's more about how to balance between the various benefits of such infrastructures and their attendant social and environmental costs.

2.3 Kenya as the Regional Fulcrum

Kenya is the current central pivot of East Africa's regional logistics network. While most of the maritime cargo into East Africa passes through the Port of Mombasa, Nairobi plays significant roles as the regions' aviation and distributive warehousing hub. This has placed the country at a strategic advantage, but also cost many negatives, more so congestion within these cities, has caused the country massive economic and environmental losses. To manage the situation, Kenya has started developing alternative hubs such as Lamu, Naivasha, and Kisumu. As

they do this, the alternative infrastructure frontend have continued to encroach and impact fragile ecosystems and local communities [16]. Studies have shown that such lean extends have also been historical through adopting green logistics practices. Studies show that measures such as eco-routing, increased use of renewable energy, and systematic carbon management can reduce environmental damage and increase competitiveness while also creating jobs [17, 18].

2.4 Thematic Literature Review by Criteria

The analysis uses five criteria from different dimensions to evaluate the location of logistics centers as follows. Environmental criteria that measure how emissions can be reduced by both increased efficiency and by capitalizing on renewable energy and effective use of land and ecosystem protection to reduce ecological degradation [6, 7, 19]. The second set of criteria, economic, compares the construction and operating costs of hub infrastructure [20] and identifies long-term benefits such as improved trade capacity and market connectivity, and job creation to support enhanced sustainable competitiveness [21, 22]. The social criteria measure the risk of displacement due to infrastructure construction, working conditions related to safety, health, diversity, equality, and inclusion, and stakeholder participation in decision-making concerning hubs [23–25]. The logistics and operational criteria give a measure of the flexibility of the hub location and readiness to support smart logistics, along with measures of multimodal connectivity, and customs and cross-border management as well as risk management to assess the system's resilience [26–28]. Finally, the strategic and geopolitical criteria assess the hub location's alignment with national development strategies, regional economic integration, the political stability, and the availability of external finance to the Five Tier Framework [29–31].

2.5 Research Gap

While the existing literature offers valuable insights about the individual aspects of sustainable logistics, most studies address different dimensions independently. This approach fails to bring an integrated view, which this study pursues. In terms of the methodology, even fewer studies apply the q -ROFS-based OPA methodology to the African context. This study aims to bridge these gaps by using a comprehensive approach that evaluates Kenya's potential logistics hubs across all five dimensions. In this way, we aim to both improve the methodology and provide practical insights for regional policy and planning.

2.6 Research Questions

RQ1. What should be the most important criteria in prioritizing sustainable logistics hubs in the East African region?

RQ2. How can the q -ROFS-based OPA method be applied to consistently weight sustainability criteria in the placement of logistics hubs in data-constrained environments?

RQ3. What is the relative importance of the identified sustainability criteria and sub-criteria in determining the strategic suitability of logistics hub locations in Kenya?

RQ4. How does a combined assessment of economic, operational, environmental, and social factors differentiate logistics hub location prioritization from efficiency-focused approaches?

RQ5. How can the results of the q -ROFS-based OPA assessment contribute to public policies related to infrastructure investments and public-private partnership strategies aimed at strengthening Kenya's role as a transit point in East Africa?

2.7 Problem Statement

Any nation that wishes to build sustainable trade competitiveness must follow a strategic approach to develop its logistics hubs. Regional trade in East Africa has seen rapid growth thanks to the EAC and the AfCFTA. This calls for an efficient logistics infrastructure that can accommodate the growing needs. Kenya is the primary maritime gateway for several of its landlocked neighbors. This means the country essentially determines how well (or poorly) regional supply chains perform, and whether they are resilient and sustainable. Kenya must expand its logistics infrastructure to accommodate the anticipated demand while balancing economic efficiency against environmental protection and social equity.

Thus far, Kenya, like other developing countries, has followed an approach in which the development of logistics hubs looks at lower costs, maximum throughput, and proximity to other transport infrastructure. Recent global shocks have shown just how narrow this approach can be, hence the need for a more inclusive approach. New approaches are those that try to understand the negative impacts of logistics hubs, such as how they contribute to greenhouse gas emissions, land-use transformation, and local environmental degradation, into the planning process. The new approach includes social equity issues such as social displacement, job insecurity, and unequal distribution of economic benefits, in addition to environmental factors.

There is a gap in the current literature on infrastructure planning. Most of the studies examine economic, environmental, social and strategic factors separately instead of looking into how they interact. This approach ignores the trade-offs that define real-life infrastructure planning and how the factors compete to define the direction taken in logistics infrastructure planning. All these point to a clear methodological and empirical gap in how sustainable logistics centers are planned in East Africa.

This study employs the use of a more integrated framework through the q -ROFS-based OPA method to prioritize sustainable logistics centers in Kenya. We aim to go beyond the limitations of common MCDM models and traditional efficiency-focused approaches. We provide a tool for supporting the development of sustainable logistics infrastructure, which balances economic objectives and environmental responsibility as well as social inclusion.

3 Methodology

The methodology section consists of the preliminaries of q -ROFSs, OPA, and the q -ROFSs-based OPA method.

3.1 Preliminaries Related to q -Ranked Orthopair Fuzzy Sets

Consider $U = \{u_1, u_2, \dots, u_n\}$ as a non-empty and fixed set. Then, a q -ROFS D on U is determined by Yager [32] as follows:

$$D = \{\langle u_i, \mu_D(u_i), v_D(u_i) \rangle \mid u_i \in U\} \quad (1)$$

where, $\mu_D(u_i)$ and $v_D(u_i)$ denote the degree of belongingness and non-belongingness for the component $u_i \in U$ on q -ROFS D respectively, and $0 \leq \mu_D(u_i), v_D(u_i) \leq 1$ with which satisfy the condition of $0 \leq (\mu_D(u_i))^q + (v_D(u_i))^q \leq 1$, where $q \geq 1$.

The degree of hesitancy of $u_i \in U$ on q -ROFS D is obtained as below:

$$H_D(u_i) = \sqrt[q]{1 - (\mu_D(u_i))^q - (v_D(u_i))^q} \quad (2)$$

where, $0 \leq H_D(u_i) \leq 1$. Yager [32] defined the q -rung orthopair fuzzy number (q -ROFN) as a pair of $\langle \mu_D(u), v_D(u) \rangle$. For convenience a q -ROFN can be represented by $D = \langle \mu_D, v_D \rangle$.

Consider $D = (\mu_D, v_D)$, $D_1 = (\mu_{D_1}, v_{D_1})$ and $D_2 = (\mu_{D_2}, v_{D_2})$ as three q -ROFNs and $\lambda > 0$. Additionally, their fundamental operations are explained as follows [32, 33].

$$D_1 \cap D_2 = \langle \min\{\mu_{D_1}, \mu_{D_2}\}, \max\{v_{D_1}, v_{D_2}\} \rangle \quad (3)$$

$$D_1 \cup D_2 = \langle \max\{\mu_{D_1}, \mu_{D_2}\}, \min\{v_{D_1}, v_{D_2}\} \rangle \quad (4)$$

$$D_1 \oplus D_2 = \left\langle \sqrt[q]{\mu_{D_1}^q + \mu_{D_2}^q - \mu_{D_1}^q \mu_{D_2}^q}, v_{D_1} v_{D_2} \right\rangle \quad (5)$$

$$D_1 \otimes D_2 = \left\langle \mu_{D_1} \mu_{D_2}, \sqrt[q]{v_{D_1}^q + v_{D_2}^q - v_{D_1}^q v_{D_2}^q} \right\rangle \quad (6)$$

$$\lambda.D = \left\langle \sqrt[q]{1 - (1 - \mu_D^q)^\lambda}, v_D^\lambda \right\rangle \quad (7)$$

$$D^\lambda = \left\langle \mu_D^\lambda, \sqrt[q]{1 - (1 - v_D^q)^\lambda} \right\rangle \quad (8)$$

Consider $D = (\mu_D, v_D)$ as a q -ROFN, different researchers determined various score functions $S(D)$ for D which are detailed as below:

(a) Yager [32] determined the $S(D)$ as follows:

$$S(D) = \mu_D^q - v_D^q \quad (9)$$

where, $-1 \leq S(D) \leq 1$.

(b) Tadić et al. [34] defined the various forms of $S(D)$ as Eq. (10):

$$S(D) = \frac{(1 + \mu_D^q - v_D^q)}{2} \quad (10)$$

where, $0 \leq S(D) \leq 1$.

(c) Wei et al. [35] obtained the different versions of $S(D)$ as follows:

$$S(D) = \frac{1}{3} (\mu_D^q - 2v_D^q - 1) + \frac{\tau}{3} (\mu_D^q + v_D^q + 2) \quad (11)$$

where, $\tau \in [0, 1]$.

The accuracy function $A(D)$ for $D = (\mu_D, v_D)$ is defined according to Yager [32] and given as follows:

$$A(D) = \mu_D^q + v_D^q \quad (12)$$

Consider $D_i = (\mu_{D_i}, v_{D_i}) \cdot (i = 1, 2, \dots, n)$ and $w = (w_1, w_2, \dots, w_n)^T$ as a weight vector related to D_i with $\sum_{i=1}^n w_i = 1$ and $w_i \in [0, 1]$. Li et al. [33] defined q -rung orthopair fuzzy weighted average (q -ROFWA) and geometric (q -ROFWG) operators as follows:

$$q\text{-ROFWA} = \left(\sqrt[q]{\left(1 - \prod_{i=1}^n (1 - \mu_{D_i}^q)^{w_i}\right)}, \prod_{i=1}^n v_{D_i}^{w_i} \right) \quad (13)$$

$$q\text{-ROFWG} = \left(\prod_{i=1}^n \mu_{D_i}^{w_i}, \sqrt[q]{\left(1 - \prod_{i=1}^n (1 - v_{D_i}^q)^{w_i}\right)} \right) \quad (14)$$

3.2 Ordinal Priority Approach

The OPA is a robust multi-criteria decision-making method developed to overcome the limitations of traditional techniques like the AHP and best–worst method [36–38]. Unlike cardinal-based approaches, OPA relies on ordinal information, namely the ranking of experts, criteria, and alternatives, to derive optimal weights [11, 38].

OPA is grounded in linear programming and does not require consistency ratios or normalization procedures, which are common sources of inconsistency in cardinal MCDM methods [37, 39]. OPA uses rank-based preferences to reduce cognitive burden and effectively accommodate incomplete information [36, 39].

The main steps of the OPA method are summarized as follows [11]:

Step 1: Identification of attributes

Key criteria are selected based on the analyst's judgment. Attributes that include sub-criteria are also incorporated into the evaluation process. Ultimately, the weights of attributes can be derived from the weights of their sub-criteria. The final level of the hierarchical decision structure (i.e., attributes) is directly engaged in the decision-making procedure.

Step 2: Selection and ranking of experts (for group decision-making)

Experts participating in the evaluation are first identified based on their expertise. They are then ranked according to factors such as organizational position, professional experience, and educational background. If two or more experts share the same rank, this equivalence is reflected in the prioritization.

Step 3: Ordering of attributes

In group decision-making, experts arrange the attributes according to their knowledge and expertise. If certain attributes are not considered critical or if experts lack sufficient knowledge to evaluate them, they may exclude those attributes from ranking and from the mathematical model. Additionally, attributes may share equal priority for a given expert, and this equivalence is incorporated into the prioritization process.

Step 4: Ranking of alternatives within each attribute

Alternatives are ranked under each attribute. In group decision-making, experts evaluate and order the alternatives according to each attribute. If some alternatives share the same priority under a specific attribute, this equivalence is taken into account in the ranking process, as expressed in Eq. (15).

$$\left(A_{ijk}^{(1)}, A_{ijk}^{(2)}, \dots, A_{ijk}^{(m)} \right) \quad (15)$$

Step 5: Solving the model, deriving attribute weights, and ranking alternatives

To determine the optimal weight of the k^{th} alternative under the j^{th} attribute at the r^{th} rank in a group decisionmaking setting, the linear mathematical model is employed. In individual decision-making, the value of (i) in the model is always equal to 1. The final weights of alternatives, attributes, and experts are obtained, as expressed in Eq. (16). Subsequently, alternatives are ranked according to these weights.

$$\left(W_{ijk}^{(1)}, W_{ijk}^{(2)}, \dots, W_{ijk}^{(m)} \right) \quad (16)$$

3.3 q -Ranked Orthopair Fuzzy Set-Based Ordinal Priority Approach Method

The q -ROFSs-based OPA method integrates ordinal ranking with fuzzy uncertainty handling [14]. The steps are as follows:

Step 1. A q -ROF-based linguistic matrix is determined. The criteria set needs to be defined in the first step related to q -ROFSs based OPA method. Consider that there are l criteria $CR_i (i = 1, 2, \dots, l)$ and f decision makers $DM_r (r = 1, 2, \dots, f)$ in the MCDM model. If criteria are divided into sub-criteria, the significance related to them needs to be determined. Following to this, decision makers need to state their judgments related to the significance of the criteria/sub-criteria with respect to the q -ROF based linguistic matrix $\mathfrak{S}^{dm} = [\xi_j^{dm}]_{l \times 1} (1 \leq dm \leq f)$:

$$\mathfrak{J}^{dm} = (\xi_1^{dm} \quad \xi_2^{dm} \quad \xi_3^{dm} \quad \dots \quad \xi_l^{dm}) \quad (17)$$

where, $\xi_j^{dm} = (\mu_j^{dm}, v_j^{dm})$ denotes the relative importance related to the criterion j that is determined by the decision maker dm via predetermined q -ROFSs based linguistic scale. After that f , q -ROFSs-based linguistic matrices are acquired.

Step 2. An aggregated q -ROFS-based linguistic matrix related to decision maker judgments is determined. The score function $S(D)$ for the relative importance in terms of the criterion $\xi_j^{dm} = (\mu_j^{dm}, v_j^{dm})$ with respect to the q -ROFSs-based linguistic matrices is obtained via Eq. (18).

$$\zeta_j^{dm} = \frac{(\mu_j^{dm})^q - 2(v_j^{dm})^q - 1}{3} + \frac{\tau}{3} \left((\mu_j^{dm})^q + (v_j^{dm})^q + 2 \right) \quad (18)$$

where, $\tau \in [0, 1]$.

The aggregated linguistic matrix related to score functions $\mathfrak{S} = [\zeta_j]_{l \times 1}$ is obtained by applying the Bonferroni function that is given by as follows:

$$\zeta_j = \left(\frac{1}{f(f-1)} \sum_{x=1}^f (\zeta_j^x)^p \sum_{\substack{y=1 \\ y \neq x}}^f (\zeta_j^y)^q \right)^{\frac{1}{p+q}} \quad (19)$$

where, the aggregated values acquired via the Bonferroni function are represented by ζ_j , stabilization parameters related to the Bonferroni function are shown by $p, q > 0$, and dm presents the dm^{th} decision maker $1 \leq dm \leq f$ [40, 41].

Step 3. According to the score function value criteria/sub-criteria are ranked. The criterion/sub-criterion with a higher ζ_j^{dm} value has a better rank. To rank the criteria/sub-criteria, the following rules need to be taken into account:

- (a) If $\zeta_j^{dm} > \zeta_{j+1}^{dm}$ then the criterion CR_j is better ranked than CR_{j+1}
- (b) If $\zeta_j^{dm} < \zeta_{j+1}^{dm}$ then the criterion CR_{j+1} is better ranked than CR_j
- (c) If $\zeta_j^{dm} = \zeta_{j+1}^{dm}$ then
 - (1) If $A_j < A_{j+1}$ then the criterion CR_{j+1} is better ranked than CR_j
 - (2) If $A_j > A_{j+1}$ then the criterion CR_j is better ranked than CR_{j+1}

where, $A_j = \mu_j^q + v_j^q$ and $A \in [0, 1]$.

The weighting coefficients related to the criterion/sub-criterion need to satisfy the condition of $wc_j^k \geq wc_j^{k+1}$, where wc_j^k denotes the weighting coefficient for the j^{th} criterion/sub-criterion having the k^{th} rank. The weighting coefficients related to successive criteria according to their rank need to meet the following condition:

$$\begin{aligned} wc_j^1 - wc_j^2 &\geq 0 \\ wc_j^2 - wc_j^3 &\geq 0 \\ &\dots \\ wc_j^{l-1} - wc_j^l &\geq 0 \end{aligned} \quad (20)$$

The aforementioned Eq. (20) can be written in the following form:

$$\frac{\min_{1 \leq j \leq l} \{\zeta_j\}}{\zeta_j} (wc_j^k - wc_j^{k+1}) \geq 0; \quad \forall_j \quad (21)$$

where, the significance related to the j^{th} criterion having the k^{th} rank is denoted by wc_j^k .

Step 4. In order to compute the weighting factors, a multi-objective nonlinear model is determined by taking the Eqs. (20) and (21) into the account.

$$\begin{aligned}
& \text{MaxMin} \left\{ \frac{\min_{1 \leq j \leq l} \{\zeta_j\}}{\zeta_j} (wc_j^k - wc_j^{k+1}); \frac{\min_{1 \leq j \leq l} \{\zeta_j\}}{\zeta_j} wc_j^n \right\}; \forall_j \\
& \text{s.t.} \\
& \sum_{j=1}^l wc_j = 1; wc_j \geq 0; \forall_j
\end{aligned} \tag{22}$$

The multi-objective nonlinear model written as Eq. (22) can be converted into a linear mathematical model given by Eq. (23):

$$\begin{aligned}
& \text{Max } \xi \\
& \text{s.t.} \\
& \frac{\min_{1 \leq j \leq l} \{\zeta_j\}}{\zeta_j} (wc_j^k - wc_j^{k+1}) \geq \xi; \forall_j \\
& \frac{\min_{1 \leq j \leq l} \{\zeta_j\}}{\zeta_j} wc_j^n \geq \xi; \forall_j \\
& \sum_{j=1}^l wc_j = 1; wc_j \geq 0; \forall_j
\end{aligned} \tag{23}$$

where, the significance related to the j^{th} criterion having the k^{th} rank is denoted by wc_j^k .

4 Case Study

A case study was performed to strategically prioritize the sustainable logistics hubs in East Africa. To determine the list of main criteria and their related sub-criteria, decision-makers from academia and industry professionals are contacted, and the extant literature is reviewed extensively. The complete list of the main criteria and sub-criteria used in the study is given in Table 1 below.

Table 1. Main criteria and sub-criteria related to study

Main Criteria (MC)	Sub-Criteria (SC)	Description	Reference
MC1. Environmental	SC1. Carbon footprint reduction potential	Ability of the logistics hub to lower greenhouse gas emissions through efficient operations and technology adoption.	[42, 43]
	SC2. Land use efficiency	Optimal utilization of available land to minimize sprawl and ecological disturbance.	[44, 45]
	SC3. Proximity to protected ecosystems	Assessment of potential risks or synergies related to biodiversity and protected environmental zones.	[19, 46]
	SC4. Waste management infrastructure	Availability and efficiency of waste collection, recycling, and sustainable disposal systems.	[47, 48]
	SC5. Renewable energy integration	Extent to which the hub incorporates renewable sources such as solar, wind, or bioenergy into operations.	[49, 50]
MC2. Economic	SC6. Infrastructure investment cost	Capital requirements for developing the logistics hub and related infrastructure.	[20, 51]
	SC7. Operational cost efficiency	Ability to minimize long-term operating costs through efficiency and innovation.	[52]

(Continued on next page)

Main Criteria (MC)	Sub-Criteria (SC)	Description	Reference
	SC8. Trade volume capacity	Potential of the hub to handle large-scale import/export flows effectively.	[21, 53]
	SC9. Connectivity to regional markets	Quality of access to regional supply chains, corridors, and key trading partners.	[54, 55]
	SC10. Job creation potential	Capacity to generate direct and indirect employment opportunities for the local and regional economy.	[22, 56]
	SC11. Local community impact	Degree to which hub development benefits surrounding communities in terms of livelihoods and services.	[57]
	SC12. Displacement or resettlement risks	Risks of forced migration or disruption of communities due to hub expansion.	[23, 58]
MC3. Social	SC13. Labor conditions and inclusivity	Fairness, safety, and inclusivity in employment practices at the logistics hub.	[24, 59]
	SC14. Stakeholder engagement	Mechanisms to ensure participation of government, private sector, and local communities in decision-making.	[25, 60]
	SC15. Public acceptance	Social legitimacy and public support for hub development and operations.	[61, 62]
MC4. Logistical Operational	SC16. Multimodal connectivity (road, rail, sea, air)	Integration of multiple transport modes for seamless freight movement.	[26]
	SC17. Customs and clearance efficiency	Speed and effectiveness of customs operations in facilitating trade flows.	[27, 63]
	SC18. Smart logistics readiness	Capacity to adopt digital and smart technologies (IoT, AI, automation).	[27]
	SC19. Scalability and future expansion	Flexibility to expand facilities and adapt to future demand.	[28, 29]
	SC20. Security and risk management	Infrastructure and processes to mitigate operational, safety, and geopolitical risks.	[30, 31]
MC5. Strategic Geopolitical	SC21. Role in regional integration (EAC, COMESA)	Contribution to strengthening Kenya's integration within regional trade blocs.	[64]
	SC22. Alignment with national development plans	Consistency with Kenya's Vision 2030, AfCFTA, and other strategic goals.	[29]
	SC23. Political stability and governance	Degree of governance reliability, security, and policy stability in supporting hub operations.	[65, 66]
	SC24. Strategic redundancy (alternative routes)	Availability of alternative logistics routes to minimize vulnerability to disruptions.	[43]
	SC25. International donor or PPP interest	Potential to attract donor support, international financing, or public-private partnerships.	[30, 31]

Four decision-makers were interviewed to assess and prioritize the main criteria and their related sub-criteria. The decision-makers are industry professionals and academicians with extensive information and experience in

sustainable logistics from Kenya-based companies and universities. Table 2 below details the information regarding the decision-makers are given in Table 2.

Table 2. Details related to decision-makers

Decision-maker (DM)	Gender	Profession	Education	Experience (Years)
DM1	Male	Senior Lecturer	PhD	14
DM2	Male	Logistics Specialist	MSc	15
DM3	Male	Lecturer	BSc	8
DM4	Female	Logistics Specialist	BSc	15

Table 3. Linguistic scale composed of q -rung orthopair fuzzy numbers (q -ROFNs) for assessing criteria/sub-criteria

Linguistic Terms	q -ROFNs for Criteria/Sub-Criteria	
	μ	ν
Absolutely low (AL)	0.15	0.95
Very low (VL)	0.25	0.85
Low (L)	0.35	0.75
Moderate low (MDL)	0.45	0.65
Moderate (MD)	0.55	0.55
Moderate high (MDH)	0.65	0.45
High (H)	0.75	0.35
Very high (VH)	0.85	0.25
Absolutely high (AH)	0.95	0.15

Table 4. Assessment of the Main Criteria (MC) and their related Sub-Criteria (SC) by decision makers (DM)

Criteria	DM1	DM2	DM3	DM4
MC1	AH	MDH	MD	H
MC2	AH	H	MD	H
MC3	H	MD	MDH	H
MC4	AH	H	MDH	MDH
MC5	VH	MDH	MDH	H
SC1	AH	VH	L	VH
SC2	H	H	MDH	AH
SC3	H	H	MDH	VH
SC4	H	H	MD	AH
SC5	H	VH	MDL	AH
SC6	AH	VH	MD	H
SC7	AH	H	MDH	AH
SC8	AH	VH	VH	H
SC9	AH	H	H	H
SC10	VH	VH	VH	AH
SC11	AH	VH	VH	AH
SC12	MDH	H	H	AH
SC13	MDH	H	H	H
SC14	MDH	H	VH	AH
SC15	H	VH	MDH	H
SC16	AH	VH	VH	MD
SC17	AH	VH	VH	H
SC18	AH	H	MDH	AH
SC19	VH	VH	VH	AH
SC20	H	VH	H	AH
SC21	AH	H	MD	AH
SC22	VH	H	VH	VH
SC23	AH	VH	AH	AH
SC24	H	H	H	VH
SC25	AH	MDH	VH	VH

Note: AH = Absolutely high; VH = Very high; H = High; MDH = Moderate high; MD = Moderate; MDL = Moderate low; L = Low.

In this study, five main criteria and a total of twenty-five sub-criteria $CR_j (j = 1, 2, \dots, 25)$, and four decisionmakers $DM_r (r = 1, 2, \dots, 4)$ were taken into account. A linguistic scale consisting of q -ROFNs was considered in prioritizing the main criteria and their related sub-criteria, as given in Table 3.

The decision-makers stated their judgments using the q -ROFNs-based linguistic scale given in Table 3. The linguistic matrices given in Table 4 show the assessment of the main criteria and their related sub-criteria outlined by the decision-makers as $\mathfrak{S}^{dm} = [\xi_j^{dm}]_{lx1} (1 \leq dm \leq 4)$.

The score functions for the MC and their SC were then obtained via Eq. (18), and presented in Table 5. The score functions of the MC and their SC were then aggregated by applying the Bonferroni function given in Eq. (19). To determine the score functions, the values $\tau = 0.8$ and $q = 1$ were considered according to the judgments of decision makers. Besides, the rank of the criteria is determined by taking the score functions as presented in the last column of the following table.

Table 5. Score functions for Main Criteria (MC) and their related Sub-Criteria (SC)

Criteria	DM1	DM2	DM3	DM4	Local	Global	Rank
MC1	0.71	0.41	0.31	0.51	0.476334	-	-
SC1	0.71	0.61	0.11	0.61	0.485414	0.231219	22
SC2	0.51	0.51	0.41	0.71	0.53078	0.252829	17
SC3	0.51	0.51	0.41	0.61	0.508103	0.242027	18
SC4	0.51	0.51	0.31	0.71	0.5023	0.239262	20
SC5	0.51	0.61	0.21	0.71	0.495629	0.236085	21
MC2	0.71	0.51	0.31	0.51	0.5023	-	-
SC6	0.71	0.61	0.31	0.51	0.526805	0.264614	15
SC7	0.71	0.51	0.41	0.71	0.579346	0.291005	8
SC8	0.71	0.61	0.61	0.51	0.608416	0.305607	5
SC9	0.71	0.51	0.51	0.51	0.557484	0.280024	12
SC10	0.61	0.61	0.61	0.71	0.634438	0.318678	3
MC3	0.51	0.31	0.41	0.51	0.431882	-	-
SC11	0.71	0.61	0.61	0.71	0.65927	0.284727	9
SC12	0.41	0.51	0.51	0.71	0.53078	0.229234	23
SC13	0.41	0.51	0.51	0.51	0.484241	0.209135	25
SC14	0.41	0.51	0.61	0.71	0.555664	0.239981	19
SC15	0.51	0.61	0.41	0.51	0.508103	0.21944	24
MC4	0.71	0.51	0.41	0.41	0.504454	-	-
SC16	0.71	0.61	0.61	0.31	0.55177	0.278343	13
SC17	0.71	0.61	0.61	0.51	0.608416	0.306918	4
SC18	0.71	0.51	0.41	0.71	0.579346	0.292254	7
SC19	0.61	0.61	0.61	0.71	0.634438	0.320045	2
SC20	0.51	0.61	0.51	0.71	0.582754	0.293973	6
MC5	0.61	0.41	0.41	0.51	0.482293	-	-
SC21	0.71	0.51	0.31	0.71	0.550097	0.265308	14
SC22	0.61	0.51	0.61	0.61	0.584373	0.281839	10
SC23	0.71	0.61	0.71	0.71	0.684466	0.330113	1
SC24	0.51	0.51	0.51	0.61	0.534335	0.257706	16
SC25	0.71	0.41	0.61	0.61	0.580972	0.280199	11

Note: DM = Decision makers.

A linear model given by Eq. (23), which was determined by applying Eqs. (20)–(22), was considered to obtain the final values of the weight coefficients related to the sub-criteria. The linear mathematical model can be written as follows:

$$\begin{aligned}
& \text{Max } \xi \\
& \text{s.t.} \\
& 0.6335 * (wc_{23} - wc_{19}) \geq \xi; \quad 0.6534 * (wc_{19} - wc_{10}) \geq \xi; \quad 0.6562 * (wc_{10} - wc_{17}) \geq \xi; \\
& 0.6814 * (wc_{17} - wc_8) \geq \xi; \quad 0.6843 * (wc_8 - wc_{20}) \geq \xi; \quad 0.7114 * (wc_{20} - wc_{18}) \geq \xi; \\
& 0.7155 * (wc_{18} - wc_7) \geq \xi; \quad 0.7186 * (wc_7 - wc_{11}) \geq \xi; \quad 0.7345 * (wc_{11} - wc_{22}) \geq \xi; \\
& 0.7420 * (wc_{22} - wc_{25}) \geq \xi; \quad 0.7463 * (wc_{25} - wc_9) \geq \xi; \quad 0.7468 * (wc_9 - wc_{16}) \geq \xi; \\
& 0.7513 * (wc_{16} - wc_{21}) \geq \xi; \quad 0.7882 * (wc_{21} - wc_6) \geq \xi; \quad 0.7903 * (wc_6 - wc_{24}) \geq \xi; \\
& 0.8115 * (wc_{24} - wc_2) \geq \xi; \quad 0.8271 * (wc_2 - wc_3) \geq \xi; \quad 0.8640 * (wc_3 - wc_{14}) \geq \xi; \\
& 0.8714 * (wc_{14} - wc_4) \geq \xi; \quad 0.8740 * (wc_4 - wc_5) \geq \xi; \quad 0.8858 * (wc_5 - wc_1) \geq \xi; \\
& 0.9044 * (wc_1 - wc_{12}) \geq \xi; \quad 0.9123 * (wc_{12} - wc_{15}) \geq \xi; \quad 0.9530 * (wc_{15} - wc_{13}) \geq \xi; \\
& 1.0000 * wc_{13} \geq \xi; \quad \sum_{j=1}^{25} wc_j = 1; wc_j \geq 0; \forall j
\end{aligned}$$

After solving the linear model, the weight coefficients related to the sub-criteria are obtained as follows:
 $w_{c_1} = 0.010858$; $w_{c_2} = 0.025638$; $w_{c_3} = 0.022550$; $w_{c_4} = 0.016663$; $w_{c_5} = 0.013741$;
 $w_{c_6} = 0.032017$; $w_{c_7} = 0.055972$; $w_{c_8} = 0.066864$; $w_{c_9} = 0.042077$; $w_{c_{10}} = 0.074505$;
 $w_{c_{11}} = 0.052418$; $w_{c_{12}} = 0.008034$; $w_{c_{13}} = 0.002554$; $w_{c_{14}} = 0.019594$; $w_{c_{15}} = 0.005234$; $w_{c_{16}} = 0.038657$;
 $w_{c_{17}} = 0.070612$; $w_{c_{18}} = 0.059542$; $w_{c_{19}} = 0.078413$; $w_{c_{20}} = 0.063132$; $w_{c_{21}} = 0.035257$;
 $w_{c_{22}} = 0.048941$; $w_{c_{23}} = 0.082445$; $w_{c_{24}} = 0.028785$; $w_{c_{25}} = 0.045499$.

According to the obtained values, while political stability and governance (SC23) was found to be the most important sub-criterion with the value of 0.082445, labor conditions and inclusivity (SC13) was acquired as the least important one with the value of 0.002554. Remaining sub-criteria are ranked as SC19 > SC10 > SC17 > SC8 > SC20 > SC18 > SC7 > SC11 > SC22 > SC25 > SC9 > SC16 > SC21 > SC6 > SC24 > SC2 > SC3 > SC14 > SC4 > SC5 > SC1 > SC12 > SC15.

4.1 Sensitivity Analysis

This study had two parameters (τ and q), whose values were determined by the decision-makers' consensus. To assess the validity and reliability of the proposed q -ROFS-based OPA methodology, the sensitivity analysis should examine the impact of different values for these two parameters on the model's initial results. Sensitivity analysis is executed in two parts. In the first part, the impact of the parameter q on the sub-criteria ranking performance is examined. Since the parameter q needs to satisfy the condition $q \geq 1$, the changes related to the parameter in the interval $1 \leq q \leq 100$ are taken into account. The results of the first part of the sensitivity analysis are depicted in Figure 1.

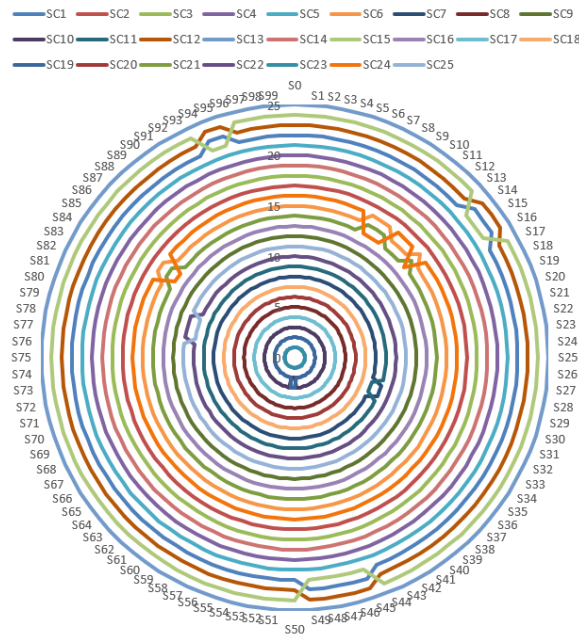


Figure 1. Ranking changes related to the sub-criteria with respect to the change of parameter q

As shown in Figure 1, small changes occurred in the ranking of sub-criteria across the scenarios created to assess the changes of parameter q . The same ranking results were obtained across 75 scenarios, with an average Spearman Rank Correlation Coefficient (SRCC) of 99%. In the second part, we assessed the impact of parameter τ from the interval $0 \leq \tau \leq 1$ on the sub-criteria ranking performance. Here, 20 scenarios were formed, with the results obtained presented in Figure 2.

As shown in Figure 2, minor changes were detected in the ranking of the sub-criteria across the scenarios created to assess changes in parameter τ . The same ranking results were returned across 11 scenarios, with an average SRCC of 99%. SC23 remained the best sub-criterion in all scenarios; SC19 was the second-best sub-criterion in 16 scenarios (80%); and SC10 was the third-best sub-criterion in 16 scenarios (80%). On the other hand, SC13 was the last sub-criterion in all scenarios. The results show that changes in parameters q and τ across various scenarios do not significantly affect the final ranking of the proposed q -ROFSs-based OPA methodology. In general, the model can be regarded as stable, valid, robust, and less sensitive to changes in parameters.

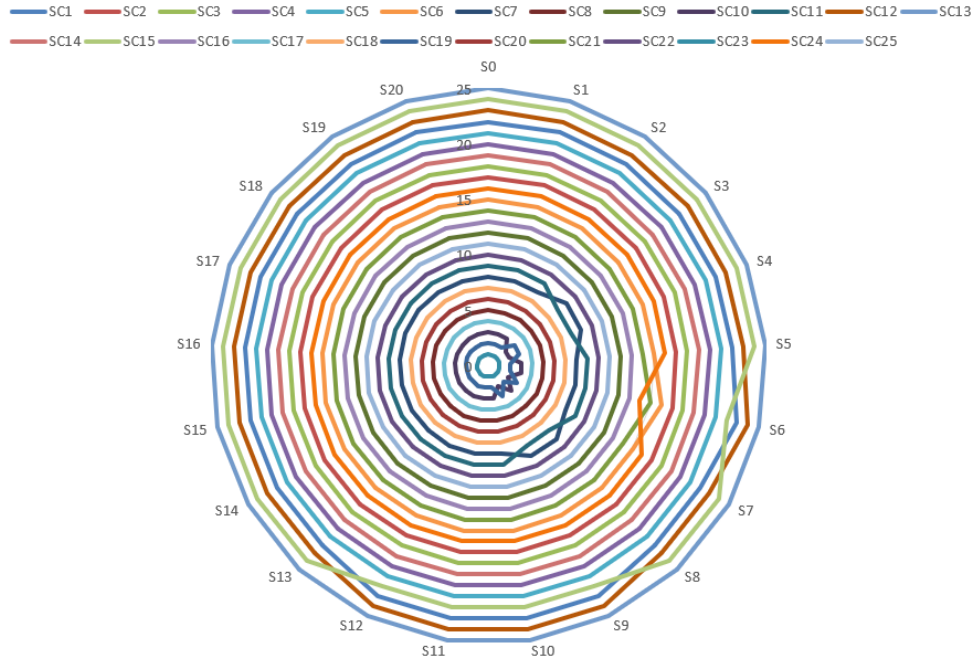


Figure 2. Ranking changes of the sub-criteria with respect to the change of parameter τ

4.2 Comparison Analysis

To assess the reliability and robustness of the ranking results from the proposed q -ROFS-based OPA, a comparative analysis was conducted using alternative fuzzy set-based OPA extensions, specifically the intuitionistic fuzzy OPA and the neutrosophic OPA. The consistency between the ranking outcomes from these methods was evaluated using the SRCC, which measures the strength and direction of monotonic relationships between ordinal rankings [67]. The resulting SRCC values are reported in Table 6, and a visual comparison of the ranking orders is shown in Figure 3.

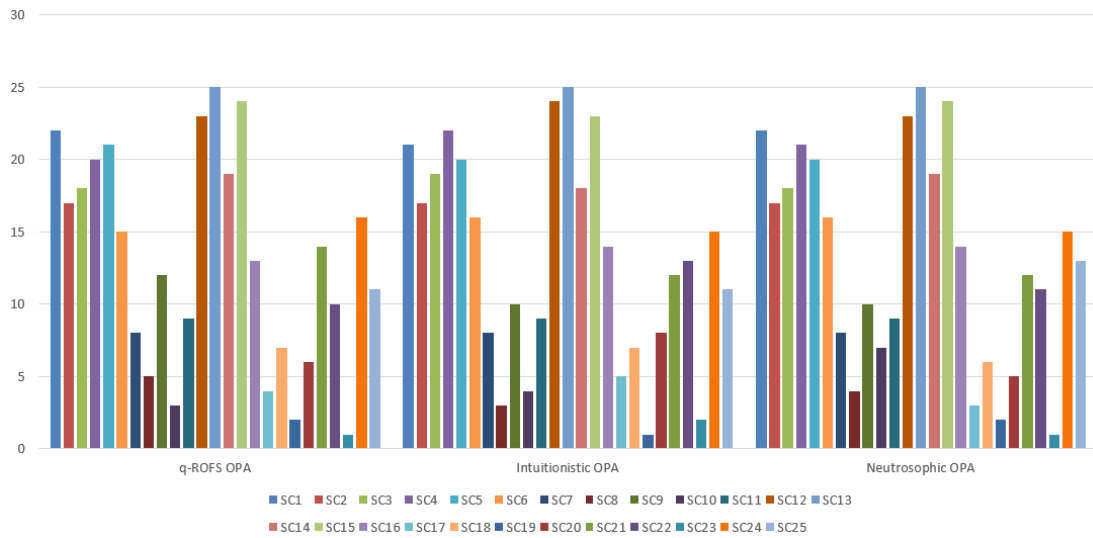


Figure 3. Ranking of sub-criteria according to various fuzzy set-based methods

As shown in Figure 3, the comparative analysis reveals slight changes in the ranking performances of the sub-criteria. According to the values in Table 6, the average SRCC value between the q -ROFS-based OPA methodology and other fuzzy sets-based techniques was 0.984. This reflects a statistically significant (at the 1% level) and strong correlation between the q -ROFS-based OPA method and other fuzzy-set-based techniques. Overall, the proposed q -ROFS-based OPA model provides consistent and dependable results compared to these techniques.

Table 6. Results of the comparison analysis

Criteria	<i>q</i> -Ranked Orthopair Fuzzy Set (<i>q</i> -ROFS) Ordinal Priority Approach (OPA)	Intuitionistic OPA	Neutrosophic OPA
SC1	22	21	22
SC2	17	17	17
SC3	18	19	18
SC4	20	22	21
SC5	21	20	20
SC6	15	16	16
SC7	8	8	8
SC8	5	3	4
SC9	12	10	10
SC10	3	4	7
SC11	9	9	9
SC12	23	24	23
SC13	25	25	25
SC14	19	18	19
SC15	24	23	24
SC16	13	14	14
SC17	4	5	3
SC18	7	7	6
SC19	2	1	2
SC20	6	8	5
SC21	14	12	12
SC22	10	13	11
SC23	1	2	1
SC24	16	15	15
SC25	11	11	13
		SRCC = 0.983	SRCC = 0.985

Note: SRCC = Spearman Rank Correlation Coefficient.

5 Conclusion

This study sought to introduce and empirically test a multi-dimensional framework that can support the strategic prioritizing of sustainable logistics hubs, with a specific focus on Kenya as a gateway to East Africa. To achieve consistent prioritization of sustainability criteria and sub-criteria, we used the *q*-ROFS-based OPA. This helped overcome the key limitations found in traditional MCDM models.

From the main sustainability dimensions, the study reveals a complex hierarchy observed among the five criteria. Logistical and operational criteria (MC4), with a local weight of 0.504454164, were found to be the most influential dimension. The implication here could be that Kenya's strategy for locating logistical hubs ranks multimodal connectivity, operational reliability, and scalability highly. Economic criteria (MC2), with a local weight of 0.502299519, ranks second. This speaks to the potential link between the economic factors to operational factors in the first position. Strategic and geopolitical criteria (MC5) (0.482293173) were ranked midway. Environmental criteria (MC1) and social criteria (MC3) at 0.476334147 and 0.431881834, respectively, seemingly play a secondary role to operational and economic considerations.

The results of the study are congruent with others in the empirical literature with respect to prioritizing connectivity and infrastructure access in determining the location of logistics hubs and dry ports. Kine et al. found proximity to road and rail networks to be the most critical criterion in their analysis of the location of an Ethiopian dry port. They concluded that access to transport corridors and intermodal feasibility hold a primary role in siting such facilities. Previous research has proposed a MCDM framework for prioritizing logistics centres, identifying existing infrastructure suitability and network efficiency as critical influencing factors. Further studies adopting a multi-dimensional assessment approach have integrated environmental, economic, infrastructural, and socio-political criteria when analysing dry port siting. Such investigations indicate that the problem is more accurately characterised as a trade-off decision rather than a single-objective optimisation task. Even with such a trade-off, the ranking of the main-criteria weights in this study leads to the conclusion that, even though a multi-dimensional approach is preferred in the prioritization of the location of logistics hubs, operational efficiency, and economic

competitiveness are still ranked higher than social or environmental, especially in regions like East Africa.

Among the sub-criteria, economic and market-oriented factors dominate the ranking. Job creation potential ranks highly, with a weight of 0.07450452 (and an overall rank of 3). Policies on the siting of infrastructure projects tend to emphasize employment generation in these developments. Trade volume capacity (0.06686421, rank 5) and operational cost efficiency (0.05597223, rank 8) also place well, further highlighting the role of scale, efficiency, and throughput. Connectivity to regional markets (0.04207657, rank 12) and the cost of infrastructure investment (0.03201686, rank 15) rank mid-level. The results show that there is more emphasis on longer-term benefits accruing from the location of these hubs vis-à-vis the short-term perspectives of cost minimization.

The study found it noteworthy that, despite their relevance to the sustainability discourse, environmental sub-criteria consistently ranked lower. Land use efficiency (0.02563788, rank 17) and proximity to protected ecosystems (0.02254997, rank 18) were ranked in the lower-middle range. Waste management infrastructure (0.016663, rank 20), renewable energy integration (0.01374079, rank 21), and carbon footprint reduction potential (0.0108575, rank 22) completed the ranking. This could imply that in the region, climate factors and the integration of green energy in infrastructure planning are merely compliance (or even suggestive) considerations rather than the primary concerns.

The study contributes to the literature on logistics and sustainability by showing that decision makers still hold economic considerations to be more important than environmental concerns. Not being ranked as the top priority does not suggest that environmental concerns are unimportant. The findings only show the reality of the context where employment creation, growth of trade opportunities, and regional competitiveness are preferred in regions such as East Africa.

The results also show the existence of a disconnect between what could be considered ambitious climate goals and the actual practice of planning and developing infrastructure. We suggest improving the regulatory environment and improving access to greenfinancing instruments as a way to fix the imbalance. With such measures, sustainability can become operational and not just remain something countries aspire to.

Future research could expand on this work in several ways. Researchers could incorporate scenario analysis or dynamic weighting to account for changing policy positions and changing priorities. A new study could take a cross-country comparative approach that integrates spatial analysis.

Author Contributions

Conceptualization, Ç.K., B.O., and M.E.; methodology, Ç.K., B.O., and M.E.; data compilation, Ç.K., B.O., and M.E.; software, Ç.K., B.O., and M.E.; validation, Ç.K., B.O., and M.E.; formal analysis, Ç.K., B.O., and M.E.; investigation, Ç.K., B.O., and M.E.; resources, Ç.K., B.O., and M.E.; writing—original draft preparation, Ç.K., B.O., and M.E.; writing—review and editing, Ç.K., B.O., and M.E.; visualization, Ç.K., B.O., and M.E. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflict of interest.

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